Changing the Perspective: model discovery in data-driven coordinates

Core technical talk

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Two main types of problems for ML application

✓ **Capturing expensive physics**

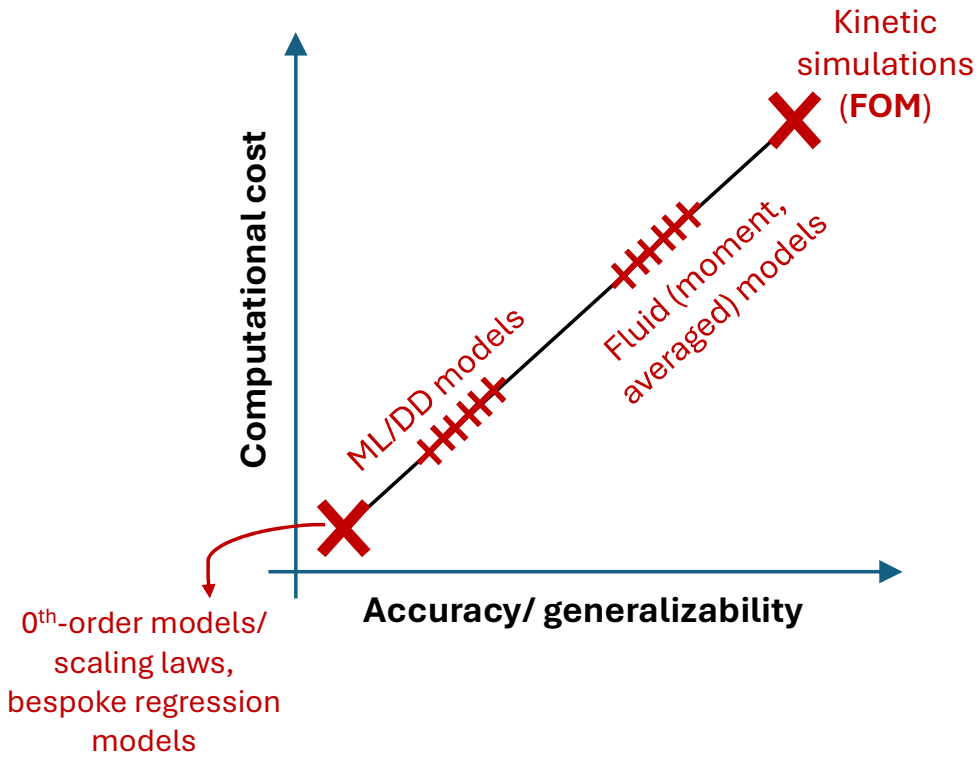
- Acceleration of high-fidelity simulations
- Reduced-order (surrogate) modelling

✓ **Dynamics discovery**

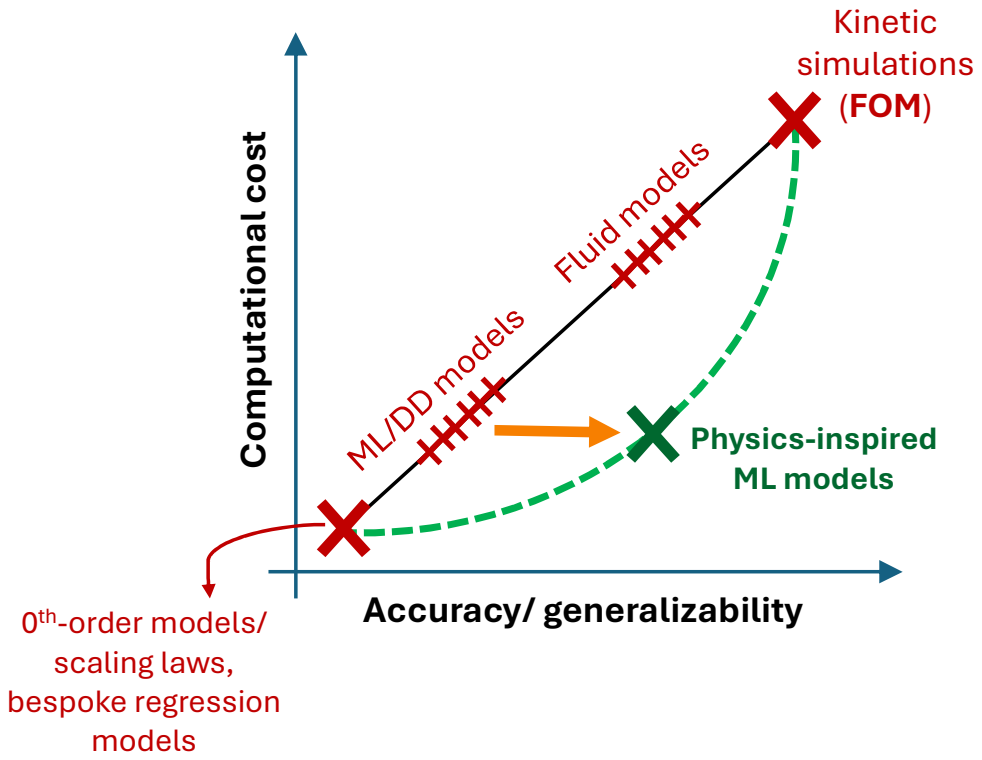
Discover **new physics** and/or **new physical correlations**, for which **governing equations** either **does not exist** or that may be **difficult to develop using first-principles**

- **Reduced-order model** (ROM) development **and dynamics discovery** are closely **related**:
  - ROMs often correspond to **discovery of evolution in transformed coordinates** (latent space), where **dynamics are not known a-priori**, hence the **need for dynamics discovery** methods.

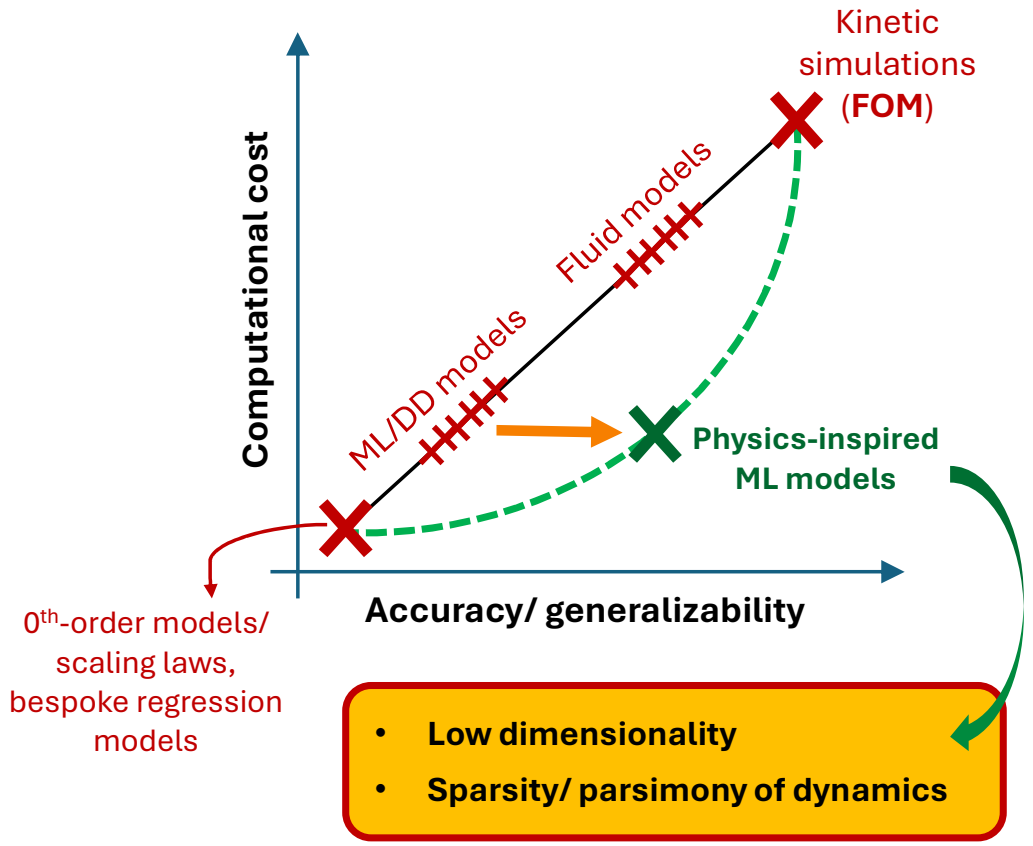
Closer look at reduced-order (surrogate) modelling



Closer look at reduced-order (surrogate) modelling



Closer look at reduced-order (surrogate) modelling



Why physics-inspired constraints?

➢ **Low dimensionality and dominant coherent structures/modes in physical systems**



Taira et al., AIAA, 2019

The **von Kármán vortex street** generated by the Rishiri island, Japan (top, photo from NASA, 2001; STS-100). This wake shares great similarity with the cylinder wake (bottom).

➢ **Sparsity (parsimony) in physics**

"Everything should be made as simple as possible, but not simpler." Albert Einstein

- **Newton's laws of motion** (classical mechanics)
- **Kepler's laws of planetary motion** (orbital mechanics)
- **Coulomb's law** (electrostatics)
- **Schrödinger's equation** (quantum mechanics) $i\hbar\frac{\partial\psi(x,t)}{\partial t} = \hat{H}\psi(x,t)$
- **Diffusion equation** (statistical mechanics) $\frac{\partial\psi}{\partial t} - \kappa\nabla^2\psi = 0$
- **Wave equation** (acoustics/electromagnetics) $-\frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} + \nabla^2\psi = 0$
- **Navier-Stokes equations** (fluid mechanics)

$$\begin{cases} \rho\frac{\partial \boldsymbol{u}}{\partial t} + +\rho(\boldsymbol{u}\cdot\nabla)u = -\nabla p + \mu\nabla^2\boldsymbol{u} + \boldsymbol{f} \\ \frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\boldsymbol{u}) = 0 \end{cases}$$

- **Maxwell's equations** (electromagnetics)

$$\begin{cases} \nabla.\boldsymbol{E} = \rho/\epsilon_0 \\ \nabla.\boldsymbol{B} = 0 \\ \nabla\times\boldsymbol{E} = -\partial\boldsymbol{B}/\partial\mathrm{t} \\ \nabla\times\boldsymbol{B} = \mu_0(+\frac{\epsilon_0\partial\boldsymbol{E}}{\partial t}) \end{cases}$$

Framework for reduced-order model development

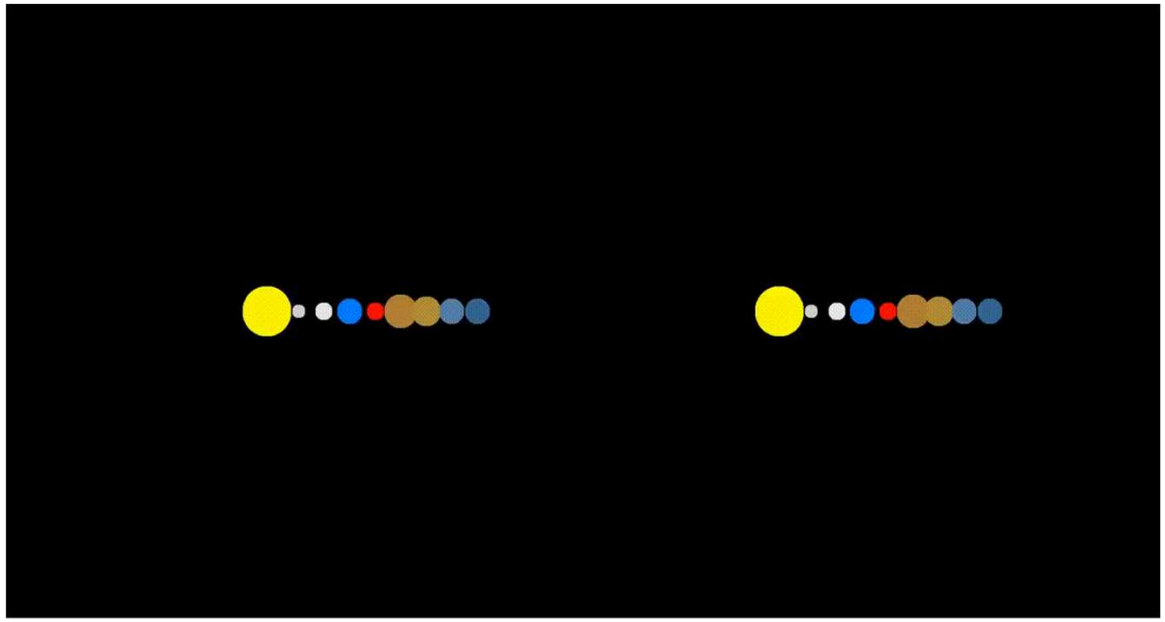
- Discovery of the intrinsic reduced coordinate system ➡ Low dimensionality
- Discovery of the dynamics in that reduced coordinate system ➡ Sparsity

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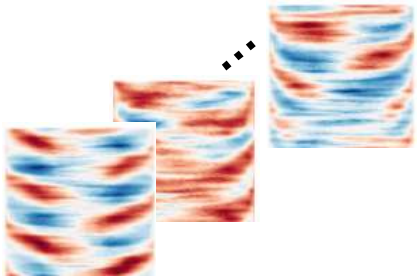


Heliocentric model
➡ Kepler's Law

Geocentric model
➡ Ptolemy's model

Two key dimensionality reduction (coordinate transformation) techniques

Singular Value Decomposition (SVD, aka POD/PCA)



- Provides a **hierarchical coordinate** system defined by **dominant correlations** within the data for the low rank representation of the data matrix
- Foundation for many ROM techniques such as dynamic mode decomposition (DMD), data compression, denoising techniques

Data Matrix

$$\mathbf{X} = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_m \\ | & | & & | \end{bmatrix}_{n \times m}, \qquad x_k \in \mathbb{R}^n$$

SVD

$$\begin{bmatrix} \\ X \\ \\ \end{bmatrix} = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_m \\ | & | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_m \end{bmatrix} \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_m \\ | & | & & | \end{bmatrix}^T$$

- **Columns of U**: left singular vectors X
- **Columns of V**: right singular vectors of X
- **Diagonal elements of Σ**: singular values,

ordered from largest to smallest

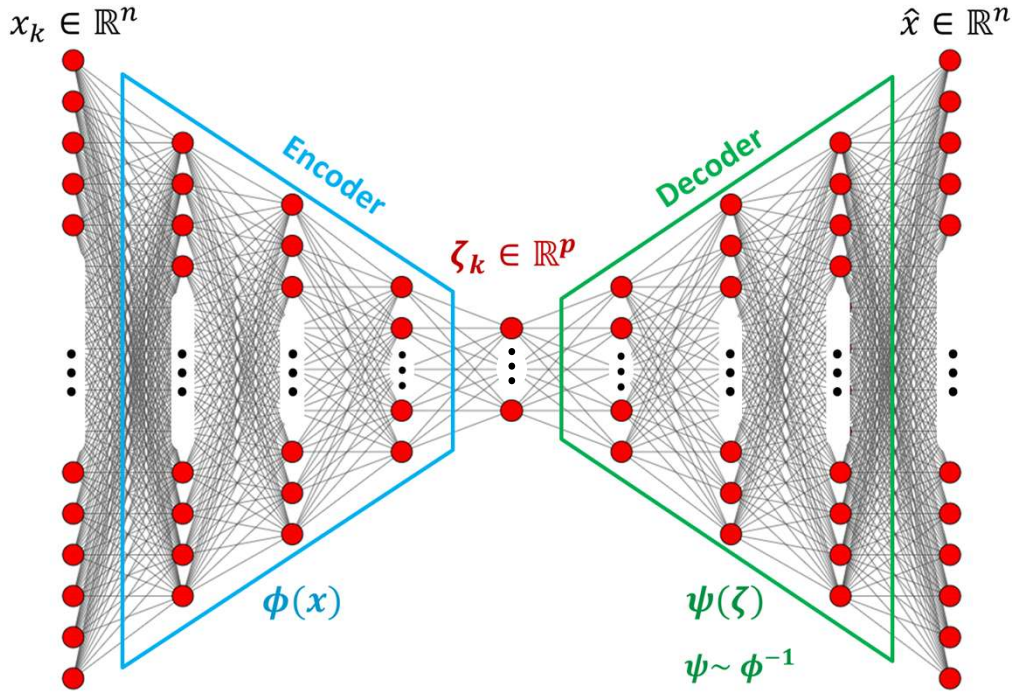
Low rank truncation

$$\tilde{X}_r = \tilde{U}_r \tilde{\Sigma}_r \tilde{V}_r{}^{\mathrm{T}} = \sum_{i=1}^{r} \sigma_i \, u_i \, v_i^T$$

Two key dimensionality reduction (coordinate transformation) techniques

Autoencoders

- By forcing as few variable as possible in the bottleneck, **distils the most important coherent structure**.
- Training autoencoder involves minimization of the loss between the reconstructed and the original data.
- Deep autoencoders generally **outperform SVD** by achieving **enhanced compression**, effectively reducing dimensionality with a smaller number of latent variables ($p \ll r$).
- **SVD** finds a **low-dimensional subspace**, while **autoencoders** discover a **nonlinear manifold** (function space) on which data exist.



Mean Squared Error

$$Loss_{MSE} = \frac{1}{m}\sum_{k=1}^{m}(\hat{x}_k - x_k)^2 \quad ; \quad \hat{x}_k = \psi(\phi(x_k;\theta_e);\theta_d)$$

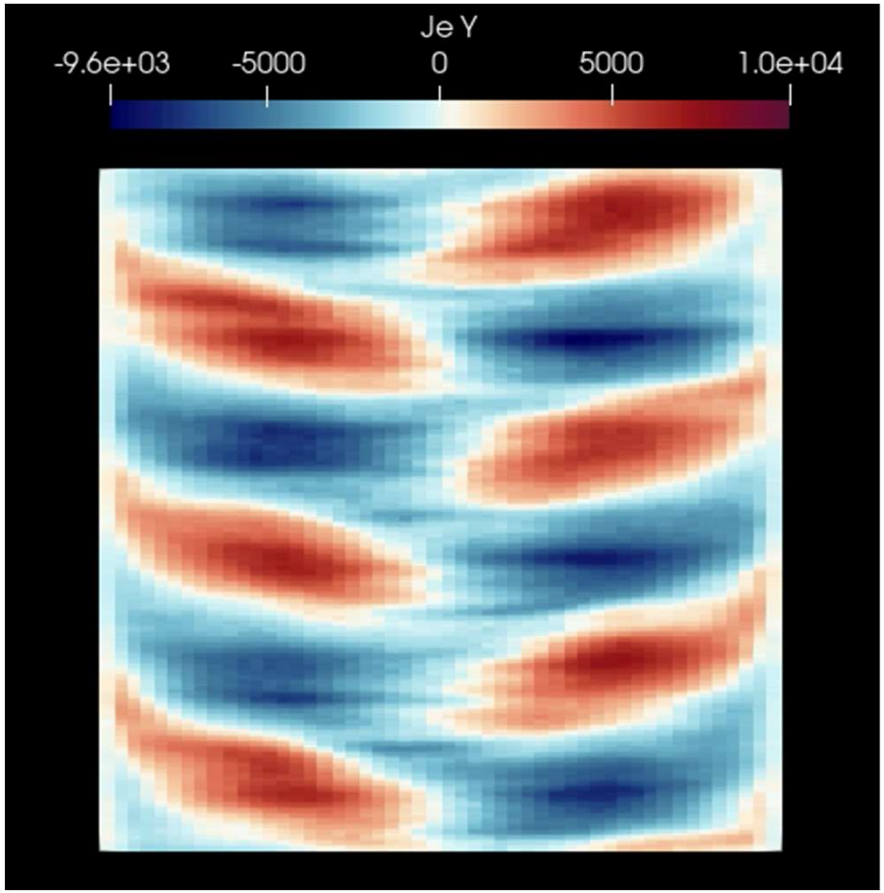
Optimization

$$\min_{\theta_e,\theta_d} \sum_{k=1}^{m}(\psi(\phi(x_k;\theta_e);\theta_d) - x_k)^2$$

θ_e, θ_d: parameters (weights) of encoder and decoder functions

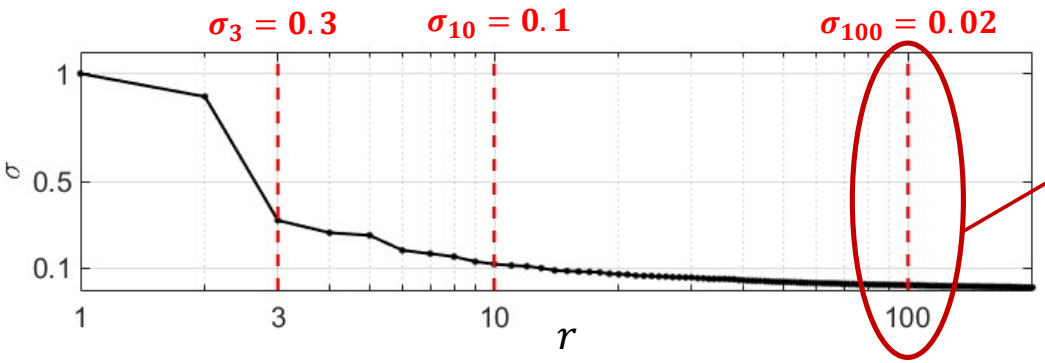
Dimensionality-reduction demonstration example for a plasma system

- **Data: Kinetic simulation of a 2D plasma discharge,** representative of radial-azimuthal cross-section of a plasma technology used for spacecraft propulsion.
- **Data matrix:** snapshots of a key physical quantity – axial electron current density (J_{ey}) – throughout time; horizontal and vertical axes represents radial and azimuthal directions, respectively.
- **Snapshot dimension:** $n = 256 \times 256 = 65{,}536$



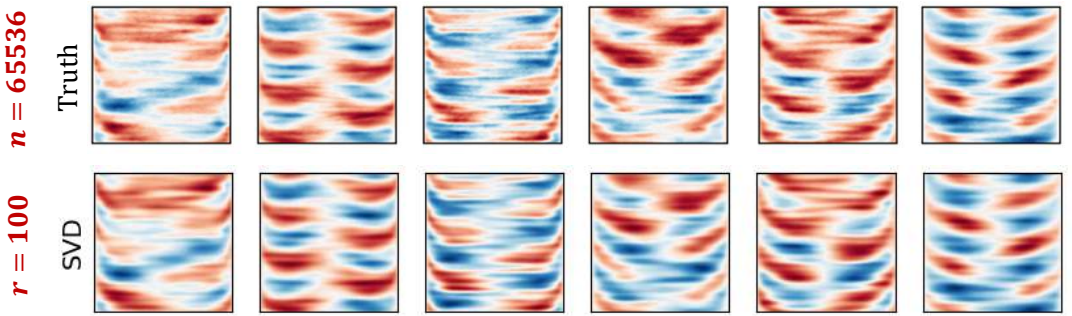
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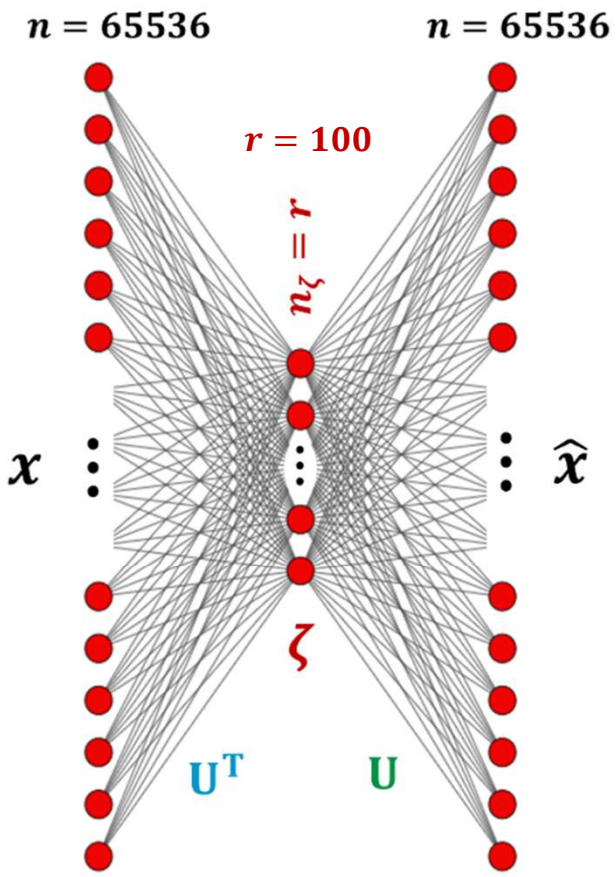
Distribution of singular values

Samples of the low rank reconstruction of 2D snapshots vs. full-rank data

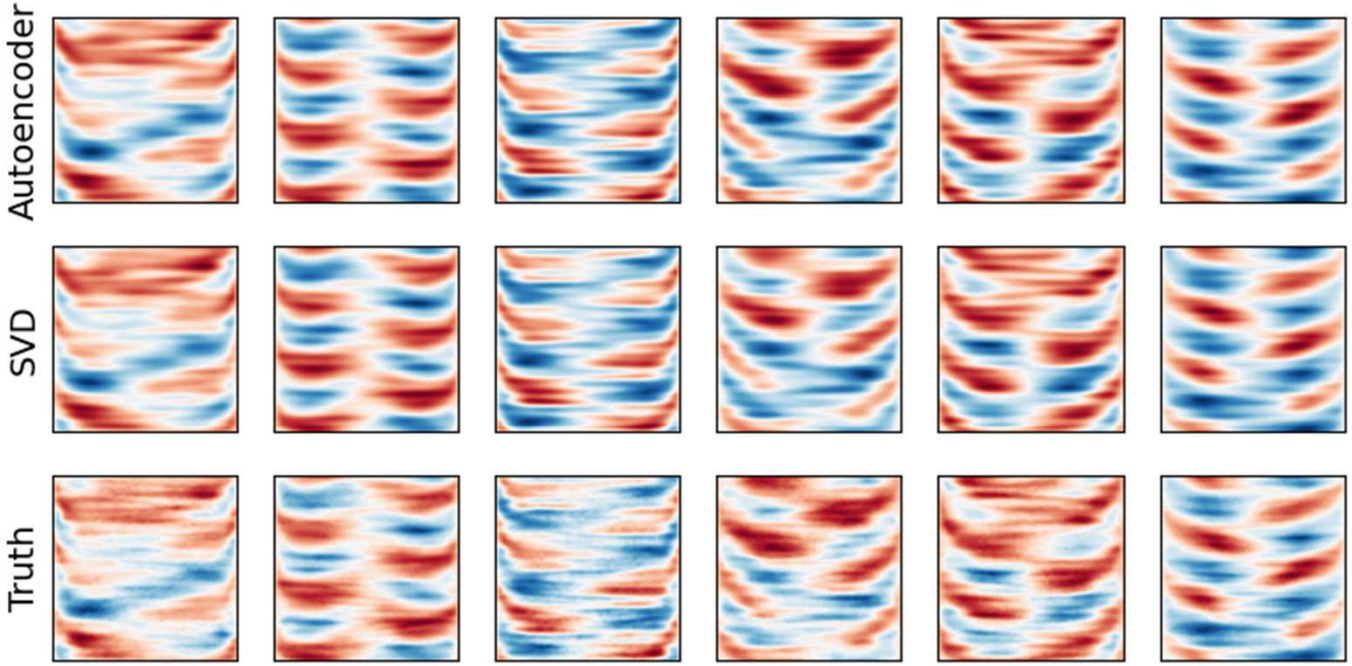


> 600 –fold dimensionality reduction

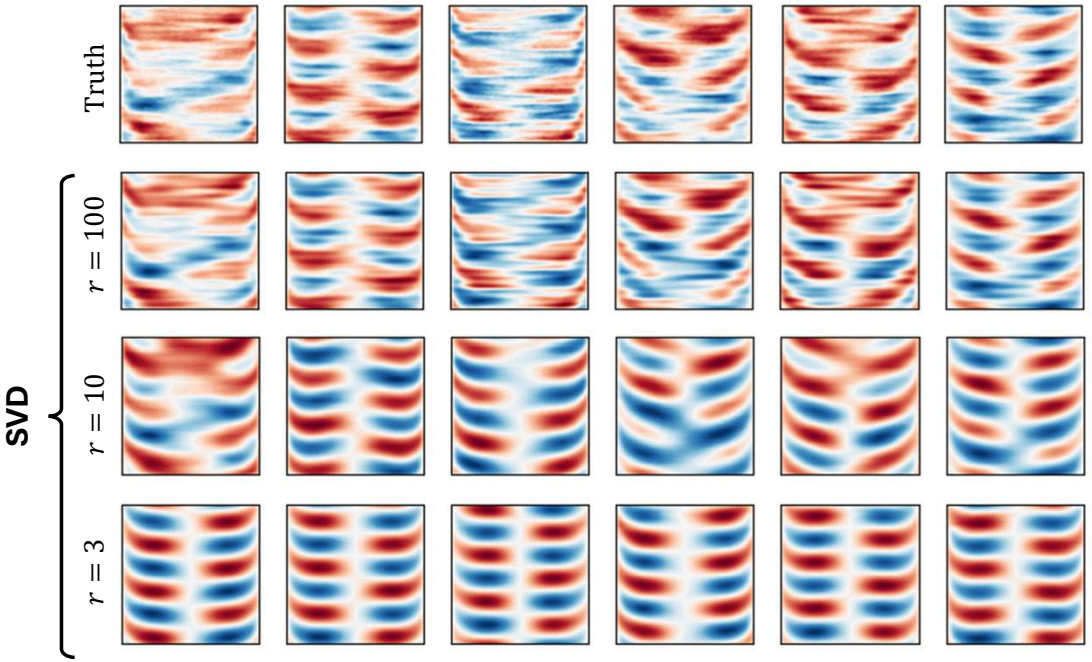
Dimensionality-reduction demonstration example for a plasma system



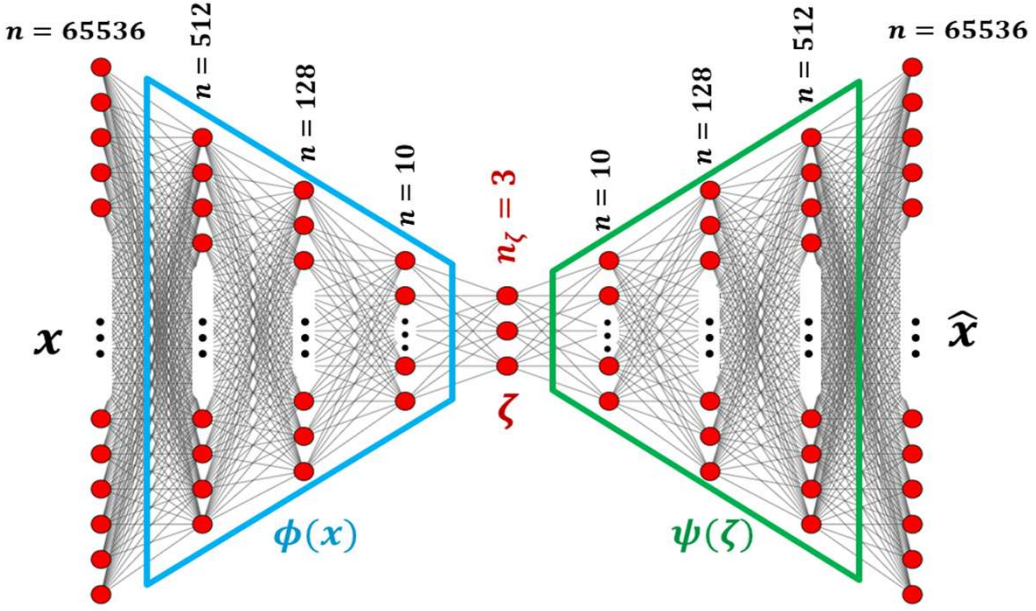
- A **rank−r SVD reconstruction** of the data is **equivalent to a linear autoencoder** with the same number r **(latent) variables** at the throat.



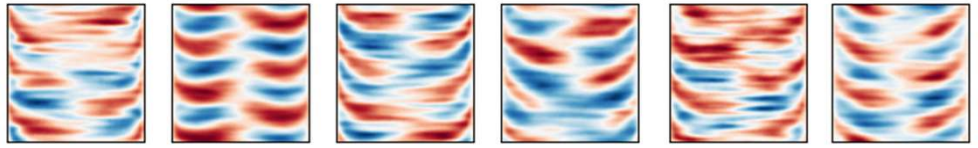
Dimensionality-reduction demonstration example for a plasma system



Samples of the low rank reconstruction of 2D snapshots vs. full-rank data



7-layer nonlinear autoencoder ($n_\zeta = 3$)



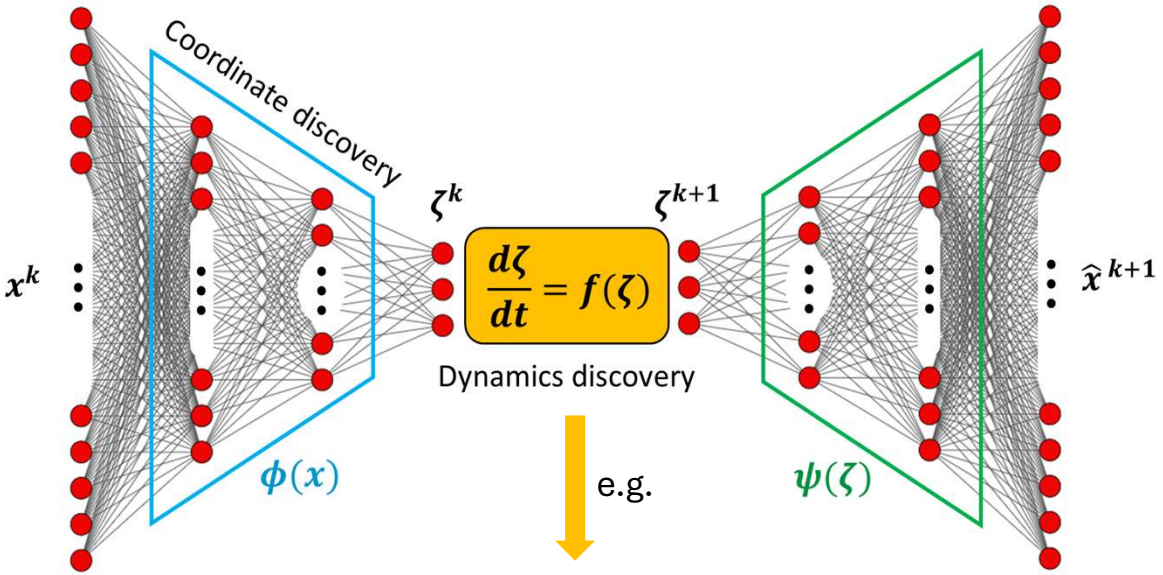
$> 20,000$ –fold dimensionality reduction

Dynamics model discovery in reduced dimension (coordinates)

- **Nonlinear coordinate transformation** using autoencoders is **an under-constrained problem** and leads to **non-unique latent representations** (the data can usually be mapped to many function spaces).

To guarantee **low dimensional** and **sparse** dynamical model (f)

Simultaneous learning of coordinate system (ζ) and dynamics ($f(\zeta)$).

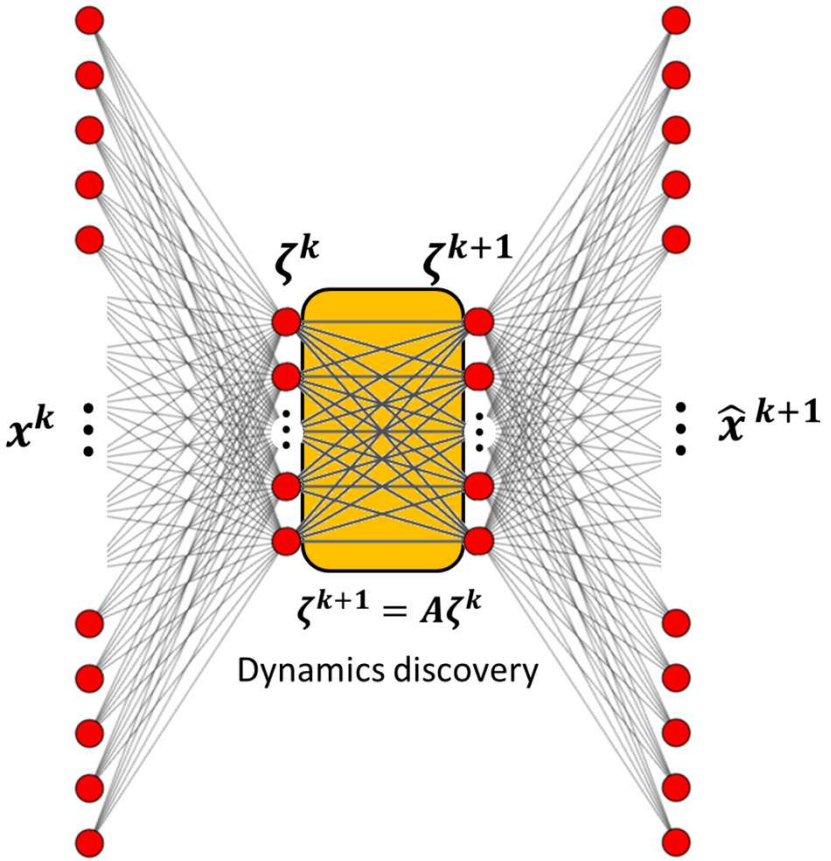


- Dynamic Mode Decomposition (DMD)
- Sparse Identification of Nonlinear Dynamics (SINDy)

Dynamics model discovery in reduced dimension (coordinates)

Optimized DMD (OPT-DMD)

- The **OPT-DMD** algorithm is a **major advance** over the Basic DMD algorithm [1][2].
- OPT-DMD **outperforms** all available DMD variants in producing **stable and consistent predictions**.
- It accounts for the **relationship among all snapshots collectively** rather than relying exclusively on pairwise connections.
- OPT-DMD **circumvents the limitation** of the Basic DMD with respect to **noise-induced bias**, making the DMD method more **stable, robust, and predictive**.

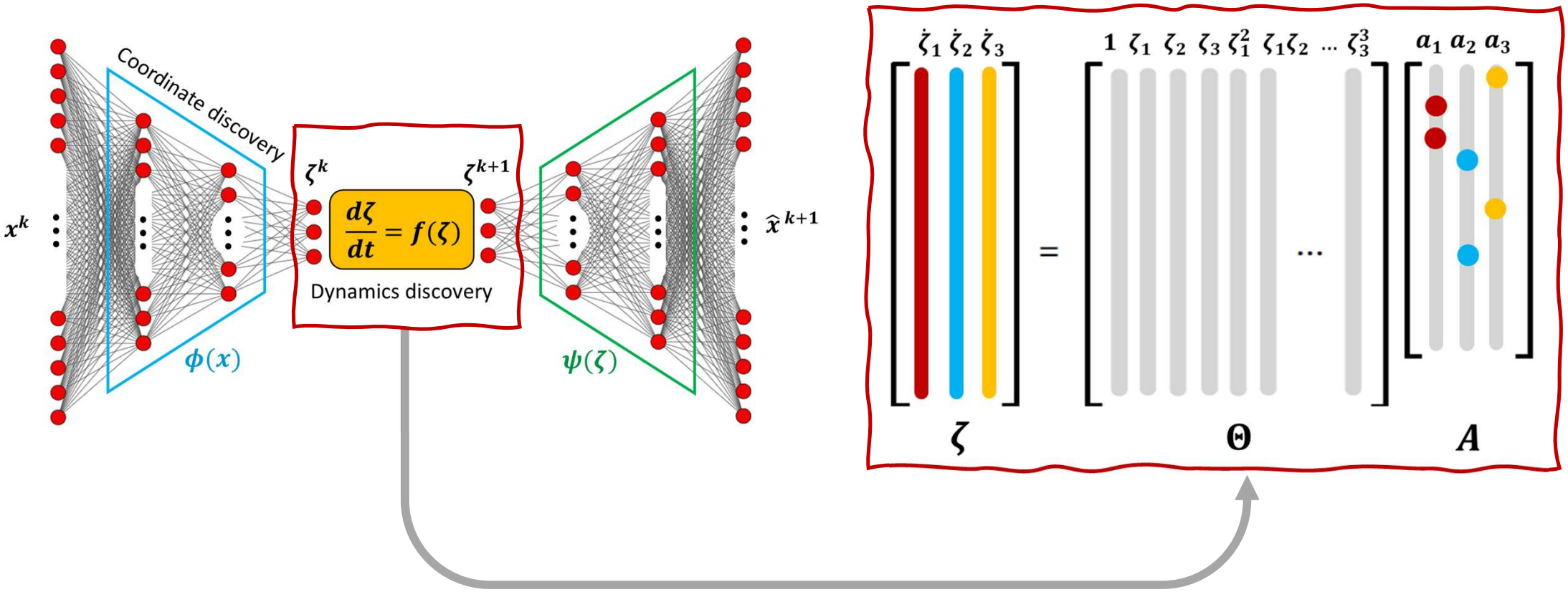


[1] Faraji F., Reza M. et al. (2024), *J.Phys. D: Appl. Phys.* 57 065202

[2] Askham T, Kutz JN, *SIAM Journal on Applied Dynamical Systems,* 17:1, 380-416 (2018)

Dynamics model discovery in reduced dimension (coordinates)

Sparse Identification of Nonlinear Dynamics (SINDy)



Dynamics model discovery with OPT-DMD

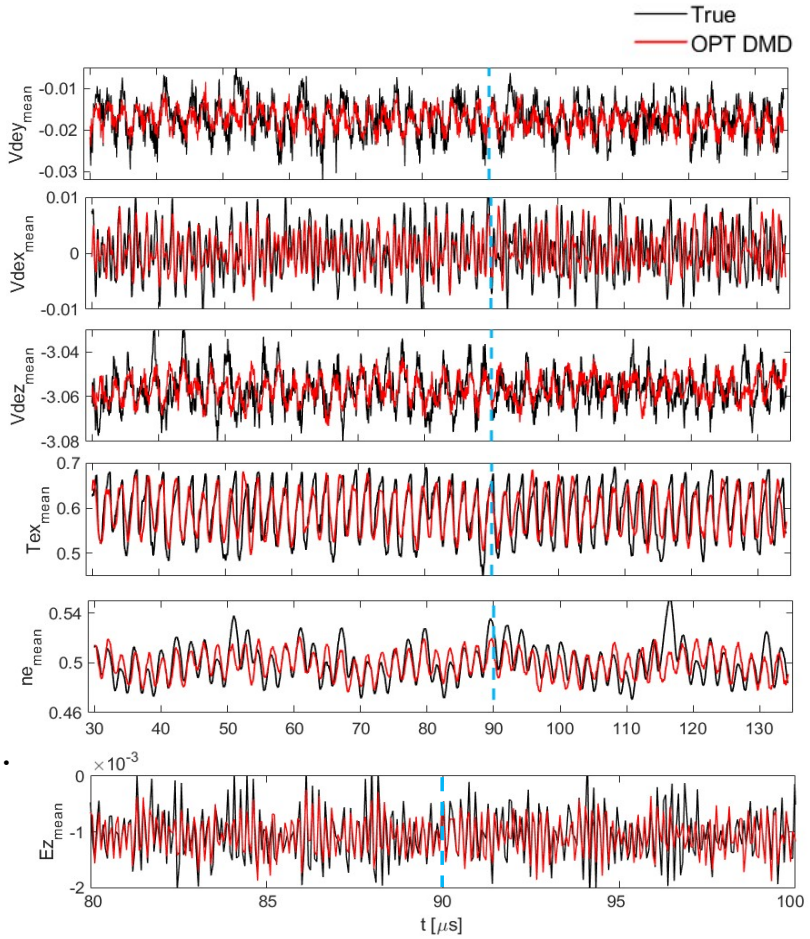
Simultaneous forecasting of a multi-property plasma state

State vector:

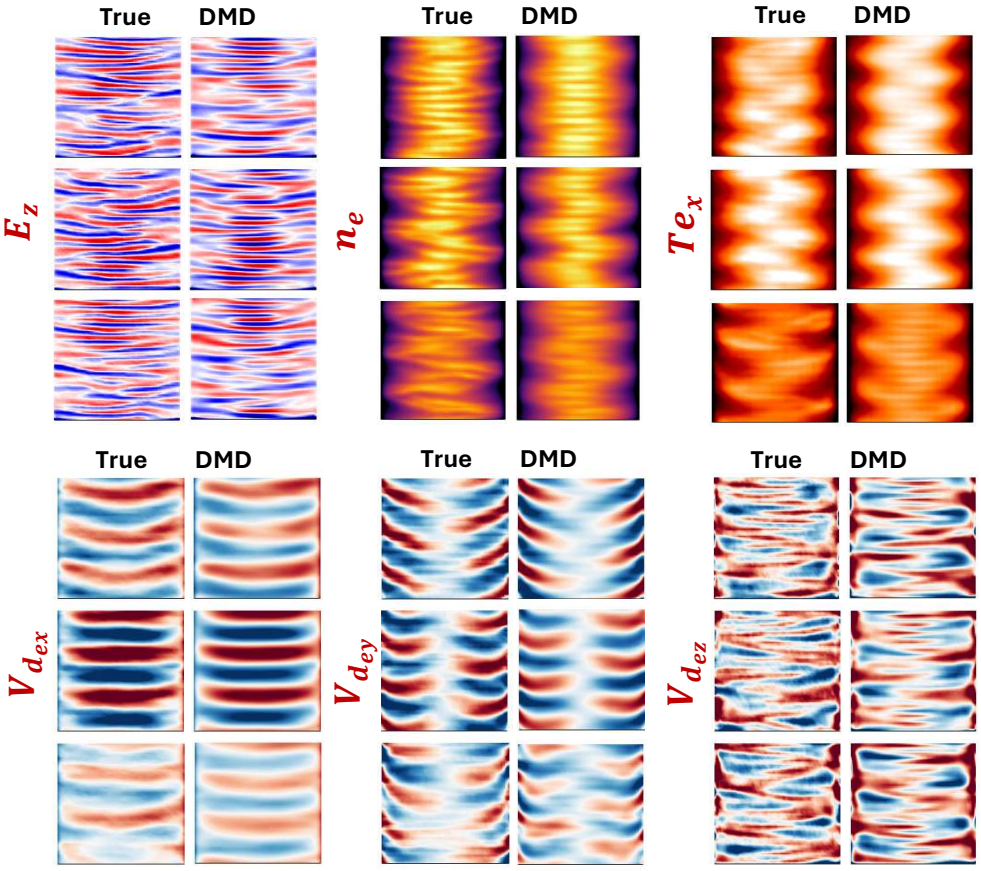
$$X = \begin{pmatrix} n_e \\ Vd_{ex} \\ Vd_{ey} \\ Vd_{ez} \\ Te_x \\ Te_z \\ E_x \\ E_z \end{pmatrix}$$

Original state vector's dim.
$n = 8 \times 65536 \sim \mathbf{5 \times 10^5}$

DMD rank = **150**



Time evolution of the spatially averaged plasma properties from OPT-DMD against ground-truth

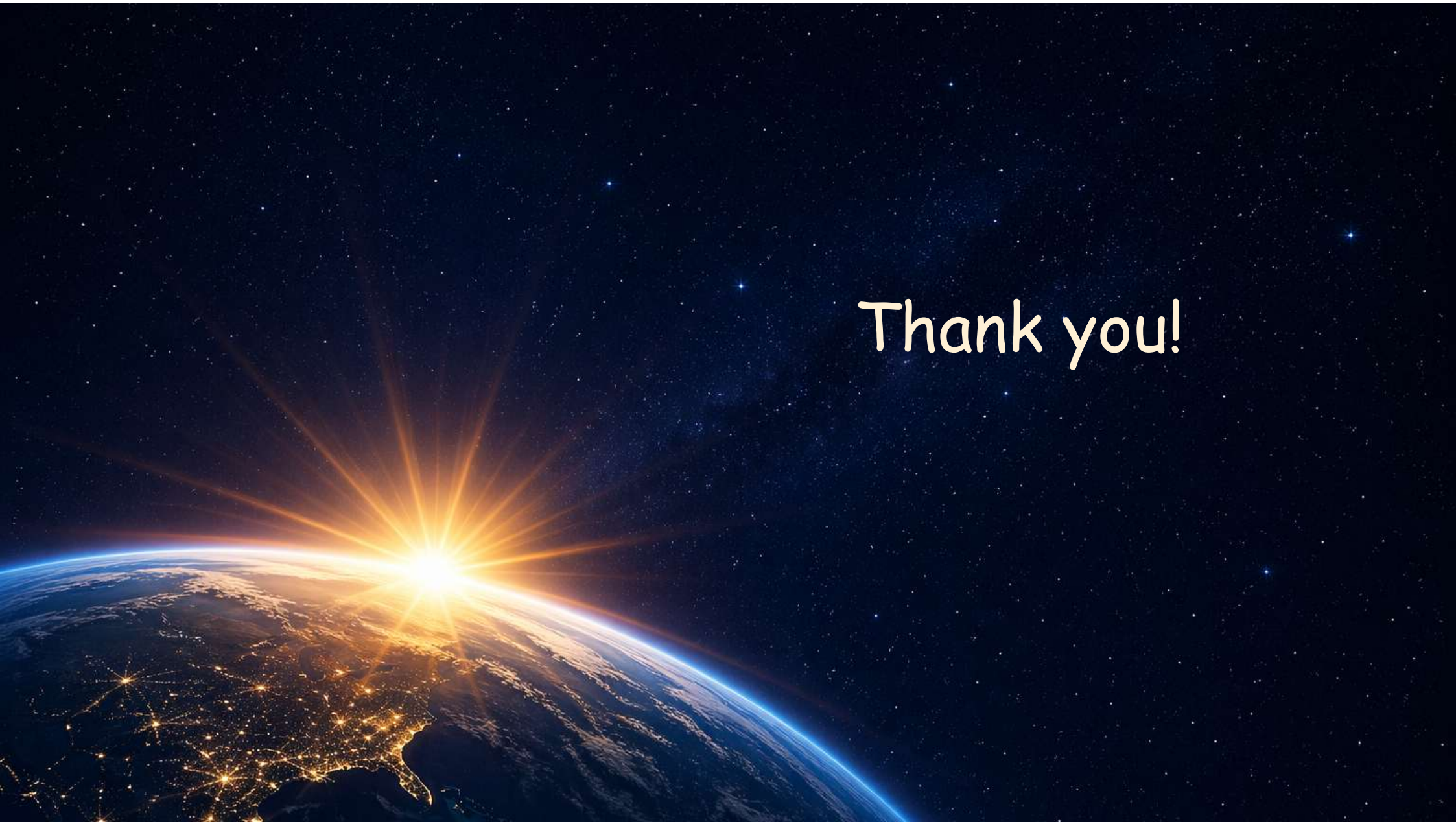


Samples of the 2D snapshots of plasma properties from OPT-DMD against ground-truth PIC simulation

The case for neuro-symbolic AI

- **Dimensionality reduction and sparsity** constraints most often do **not fully resolve** the problem of **identifying a unique solution** that respects fundamental physics constraints (symmetries, invariances, conservations) and can generalize beyond training data distribution.

- **Logic-based enforcement** of physics constraints within the neuro-symbolic AI frameworks can **mitigate the identifiability problem** by guiding coordinate-model discovery through **symbolic reasoning toward a physics-consistent, generalizable solution**.


Thank you!